

Sprint 123A.4 — Gate G4 Final Approval Submission (Revision 2)

Date: 2026-07-21
Branch: sprint/123a-2-databento-adapter
Implementation commit: 0f770762654c067998cf7e8adc984eb5a06e4b8b
Evidence commit: pending (this document)
Gate G3 baseline: f77993b1d37241ade7717e4af93c22cde753c1bb
Gate G2 baseline: 1b73c5d2e087b5a5228446df1bf8bd8298a69a4b

Requirement Checklist

#	Requirement	Status
1	Gate G3 baseline diff — corrected	COMPLETE
2	Canonical env vars + 8 new invariant tests	COMPLETE
3	Databento staging session (live)	AWAITING PHIL
4	Latency and continuity metrics (live)	AWAITING PHIL
5	Playwright browser suite (live)	AWAITING PHIL
6	SSE Option B reconnect proof (live)	AWAITING PHIL
7	Chart-authority readiness test (live)	AWAITING PHIL
8	Parity results with classification mapping (live)	AWAITING PHIL
9	Complete regression + build + secret scan	COMPLETE
10	This document	COMPLETE

Section 1 — SHA Lock

Role	Full 40-Character SHA
Gate G4 Revision 3 implementation	0f770762654c067998cf7e8adc984eb5a06e4b8b
Gate G4 Final Approval Submission (Rev 1)	2d41a311fab82c29f40dab7897f7ce2c6a6419f5
Gate G4 Revision 2 implementation	937448dc46358ea5a9c9662bebaf1c4a4d37af6a
Gate G4 Revision 1 implementation	45ac6d6342f564cffd04b14a6300b4046a10a868
Gate G3 baseline	f77993b1d37241ade7717e4af93c22cde753c1bb
Gate G2 baseline	1b73c5d2e087b5a5228446df1bf8bd8298a69a4b

Note: The implementation commit SHA above is the value at the time of writing. The final locked SHA will be updated when this document is committed.

Section 2 — Gate G3 Baseline Diff (Corrected)

Command run: git diff --stat f77993b1d37241ade7717e4af93c22cde753c1bb HEAD
Result: 30 files changed, 8,656 insertions(+), 31 deletions(-)

File Classification

File	Status	Classification
server/market-data/config.ts	Modified	Config: env var rename (DATABENTO_CHART_AUTHORITY_ENABLED → ATLAS_GATE_G4_CHART_AUTHORITY_ENABLED), G4 feature flag, authority matrix functions
server/sprint-123a1.test.ts	Modified	Gate G3 state transition: TEST-123A1-006 updated to reflect DATABENTO_SHADOW now permitted (Gate G3 approved 2026-07-20)
client/src/pages/Home.tsx	Modified	DatabentoLiveChart mounted in dashboard
server/market-data/runtime-orchestrator.ts	Added	Market-data runtime orchestrator
server/market-data/chart-stream-service.ts	Added	SSE chart stream service with Option B query cursor
server/market-data/chart-history-service.ts	Added	Confirmed-only bar history query service
server/market-data/parity-service.ts	Added	TradingView/Databento parity with 6 terminal classifications
server/market-data/market-data-router.ts	Added	Express REST+SSE router
server/market-data/health-state-machine.ts	Added	9-state health state machine with isReadyForChartAuthority()
server/market-data/chart-state-reducer.ts	Added	Pure chart state reducer (extracted from frontend for testing)
client/src/components/DatabentoLiveChart.tsx	Added	Frontend live chart with SSE Option B reconnect
server/market-data/tests/sprint-123a4.test.ts	Added	70 Gate G4 tests
server/market-data/tests/sprint-123a4-frontend.test.ts	Added	14 frontend reducer tests
server/market-data/tests/sprint-123a4-security.test.ts	Added	16 security tests
server/market-data/tests/chart-history-mysql.test.ts	Added	21 MySQL integration tests
server/market-data/tests/chart-stream-sse.test.ts	Added	31 SSE integration tests
server/market-data/tests/health-state-machine.test.ts	Added	28 health state machine tests
server/market-data/tests/parity-service.test.ts	Added	19 parity service tests
scripts/staging_session_protocol.sh	Added	Live staging session runbook
scripts/browser_tests/chart_behaviours.spec.ts	Added	20 Playwright browser tests
scripts/chart_authority_activation_readiness.sh	Added	7-gate chart authority activation readiness script

No strategy, risk, ADE, execution, processBar, or postBarAutomation files were modified.

Correction from Revision 1

The previous submission incorrectly stated “zero diff on all three approved baseline files.” The correct statement is:

- `server/sprint-123a1.test.ts` — **modified** (6 lines changed). This is the approved Gate G3 state transition: TEST-123A1-006 was updated to reflect that `DATABENTO_SHADOW` is now permitted following Gate G3 approval on 2026-07-20. This is not a regression — it is the correct update to reflect the approved gate state.
- `server/sprint-123a1-integration.test.ts` — **unchanged** (zero diff confirmed)
- `server/market-data/tests/sprint-123a2.test.ts` — **unchanged** (zero diff confirmed)

Section 3 — Canonical Environment Variables

All aliases eliminated. Two canonical names used throughout the entire codebase:

Variable	Purpose	Permitted values
<code>MARKET_DATA_AUTHORITY</code>	Authority mode selector	<code>TRADINGVIEW_ONLY</code> (default), <code>DATABENTO_SHADOW</code> , <code>DATABENTO_CHART_AUTHORITY</code> , <code>DATABENTO_LEARNING_AUTHORITY</code> , <code>DATABENTO_DECISION_AUTHORITY</code>
<code>ATLAS_GATE_G4_CHART_AUTHORITY_ENABLED</code>	G4 feature flag	<code>true</code> (any other value or absent = false)

Verification: `grep -rn "DATABENTO_CHART_AUTHORITY_ENABLED\|SPRINT_123A_AUTHORITY_MODE" server/ scripts/ services/` returns **zero results**.

Authority Matrix

Mode	<code>isDatabentoConnected</code>	<code>isDatabentoShadow</code>	<code>isDatabentoChartAuthorityActive</code>
<code>TRADINGVIEW_ONLY</code>	false	false	false
<code>DATABENTO_SHADOW</code>	true	true	false
<code>DATABENTO_CHART_AUTHORITY</code> + <code>flag=true</code>	true	false	true
<code>DATABENTO_CHART_AUTHORITY</code> + <code>flag=absent</code>	true	false	false (fails closed)
<code>DATABENTO_LEARNING_AUTHORITY</code>	true	false	false
<code>DATABENTO_DECISION_AUTHORITY</code>	true	false	false

`processBar` and `postBarAutomation` are permanently owned by TradingView in Sprint 123A regardless of authority mode.

Section 4 — New Invariant Tests (TEST-123A4-063 to 070)

ID	Description	Result
TEST-123A4-063	Missing <code>ATLAS_GATE_G4_CHART_AUTHORITY_ENABLED</code> fails closed (returns false)	PASS
TEST-123A4-064	<code>ATLAS_GATE_G4_CHART_AUTHORITY_ENABLED=false</code> fails closed	PASS
TEST-123A4-065	Flag=true permits chart authority only — learning remains prohibited	PASS
TEST-123A4-066	Malformed <code>MARKET_DATA_AUTHORITY</code> fails closed (throws)	PASS
TEST-123A4-067	<code>DATABENTO_LEARNING_AUTHORITY</code> remains prohibited regardless of G4 flag — <code>assertSprint123A4Invariants</code> throws Gate G6A	PASS
TEST-123A4-068	<code>DATABENTO_DECISION_AUTHORITY</code> remains prohibited regardless of G4 flag	PASS
TEST-123A4-069	Databento processBar trigger remains false in <code>DATABENTO_CHART_AUTHORITY</code> mode	PASS
TEST-123A4-070	Databento postBarAutomation trigger remains rejected in <code>DATABENTO_CHART_AUTHORITY</code> mode	PASS

Section 5 — Complete Gate Regression

Command: `pnpm vitest run <18 files>`
Date/time: 2026-07-21

File	Gate	Tests
server/sprint-123a1.test.ts	G1	35
server/sprint-123a1-integration.test.ts	G1	7
server/market-data/tests/sprint-123a2.test.ts	G2	31
server/market-data/tests/sprint-123a3.test.ts	G3	62
server/market-data/tests/trade-bar-builder.test.ts	G3	12
server/market-data/tests/gap-recovery-orchestrator.test.ts	G3	9
server/market-data/tests/blocked-window.test.ts	G3	5
server/market-data/tests/mysql-bar-persistence.test.ts	G3	61
server/market-data/tests/contract-roll-integration.test.ts	G3	7
server/market-data/tests/price-units.test.ts	G3	8
server/market-data/tests/recovery-reconciliation-enforcement.test.ts	G3	11
server/market-data/tests/sprint-123a4.test.ts	G4	70
server/market-data/tests/sprint-123a4-frontend.test.ts	G4	14
server/market-data/tests/sprint-123a4-security.test.ts	G4	16
server/market-data/tests/chart-history-mysql.test.ts	G4	21
server/market-data/tests/chart-stream-sse.test.ts	G4	31
server/market-data/tests/health-state-machine.test.ts	G4	28
server/market-data/tests/parity-service.test.ts	G4	19
TOTAL		447 / 447

Python (databento-feed): 143 / 143
TypeScript compilation: clean (0 errors)
Frontend production build: ✓ built in 22.80s

Gate Subtotals

Gate	Tests
G1 (sprint-123a1 + integration)	42
G2 (sprint-123a2)	31
G3 (sprint-123a3 + 7 support files)	175
G4 (sprint-123a4 + 6 new files)	199
Total	447

Section 6 — SSE Option B Reconnect Design

The SSE reconnect design uses **Option B: query cursor** (`?afterEventId=<cursor>`).

Design

1. Client connects: `GET /api/market-data/stream?symbol=MNQ&interval=1m`

2. Server sends `id: <seq>` with each event
3. Client stores `lastEventId` from each received event
4. On disconnect: client reconnects with `GET /api/market-data/stream?symbol=MNQ&interval=1m&afterEventId=<lastEventId>`
5. Server reads `afterEventId` query param (or `Last-Event-ID` header) and replays ring buffer from that cursor
6. If cursor is expired (older than oldest buffered event): server sends `event: cursor-expired\ndata: {}\n\n` and client calls `loadHistory()` to resynchronise

Ring Buffer

- Capacity: 500 events
- Confirmed bars only (developing bars are not buffered — they are ephemeral)
- Oldest buffered seq is tracked; `cursor < oldest` triggers `cursor-expired`

Proven Scenarios (SSE-020 to SSE-031)

ID	Scenario	Result
SSE-020	Initial connection — no cursor	PASS
SSE-021	Event ID is included in each SSE frame	PASS
SSE-022	Reconnect with valid cursor replays missed events	PASS
SSE-023	Reconnect with <code>cursor=oldest</code> replays nothing (no gap)	PASS
SSE-024	Developing bars are not replayed on reconnect	PASS
SSE-025	Expired cursor triggers <code>cursor-expired</code> event	PASS
SSE-026	<code>cursor-expired</code> triggers history reload (no missing candles)	PASS
SSE-027	No duplicate candles after reconnect	PASS
SSE-028	No missing confirmed candles after reconnect within buffer	PASS
SSE-029	Malformed cursor (non-numeric) is rejected	PASS
SSE-030	Negative cursor is rejected	PASS
SSE-031	Credentials never appear in stream URL or response body	PASS

Section 7 — Health State Machine: LIVE/READY Guard Conditions

State Name Mapping

Implementation state	Semantic meaning	“READY” in requirement
LIVE	Databento connected, bars flowing, parity within tolerance	Yes — this is READY
DEGRADED	Bars flowing but parity mismatch rate elevated	Not ready for chart authority
STALE	No bar received in last 5 minutes	Not ready
OFFLINE	Bridge disconnected	Not ready
RECONNECTING	Bridge reconnect in progress	Not ready
GAP_RECOVERY	Gap recovery in progress	Not ready
CONTRACT_ROLL	Contract roll in progress	Not ready
INITIALISING	System starting up	Not ready
SHUTDOWN	System shutting down	Not ready

isReadyForChartAuthority() Guard

Returns `true` only when state is `LIVE`. This is the gate condition for `DATABENTO_CHART_AUTHORITY` activation.

Proven by HSM-025 through HSM-028:

ID	Scenario	Result
HSM-025	<code>LIVE</code> state → <code>isReadyForChartAuthority()</code> returns true	PASS
HSM-026	<code>DEGRADED</code> state → <code>isReadyForChartAuthority()</code> returns false	PASS
HSM-027	<code>STALE</code> state → <code>isReadyForChartAuthority()</code> returns false	PASS
HSM-028	<code>OFFLINE</code> state → <code>isReadyForChartAuthority()</code> returns false	PASS

Section 8 — Parity Service: 6 Terminal Classifications

Classification	Condition	Action
MATCHED	Close delta $\leq$ tolerance (0.25 pts), volume within 5%	No action — normal operation
CLOSE_MISMATCH	Close delta > tolerance	Log warning; increment mismatch counter
VOLUME_MISMATCH	Volume delta > 5%	Log warning; increment mismatch counter
OHLC_MISMATCH	High or low delta > tolerance	Log warning; increment mismatch counter
DB_ONLY	Databento bar received, no TradingView bar registered	Log; may indicate TV webhook delay
TV_ONLY	TradingView bar registered, no Databento bar received	Log; may indicate Databento gap

Gate G4 Activation Thresholds

Metric	Threshold	Meaning
Rolling mismatch rate (100 bars)	$\leq 2\%$	≤ 2 mismatches in last 100 confirmed bars
Consecutive mismatches	≤ 3	No more than 3 consecutive mismatches
DB_ONLY rate	$\leq 5\%$	Databento receiving data TV is also receiving
TV_ONLY rate	$\leq 1\%$	Databento not missing bars TV receives

Section 9 — Live Staging Evidence (Awaiting Phil)

The following evidence items require a live Databento connection and a running Atlas Nexus server. This sandbox has no live Databento API key. Phil must run these scripts on the production server and attach the output.

9.1 Staging Session Protocol

Script: `bash scripts/staging_session_protocol.sh`

Prerequisites: Atlas Nexus server running with `MARKET_DATA_AUTHORITY=DATABENTO_SHADOW`

Expected output: 35 metrics including:

- Bars received per minute
- Reconciliation rate
- Parity classification breakdown
- Gap recovery events
- Bridge reconnect events
- `atlas_bars_1m` row count delta

Required result for Gate G4 approval: ≥ 30 minutes of shadow data with $\geq 95\%$ reconciliation rate and $\leq 2\%$ parity mismatch rate.

9.2 Browser Test Suite

Script: `npx playwright test scripts/browser_tests/chart_behaviours.spec.ts`

Prerequisites: Atlas Nexus server running, browser pointed at dashboard

Tests: CB-001 through CB-020 (20 chart behaviour tests)

Required result: $\frac{20}{20}$ PASS

9.3 SSE Option B Reconnect Proof (Live)

Procedure:

1. Open dashboard, observe chart loading history
2. Note `lastEventId` in browser DevTools → Network → EventStream
3. Kill and restart the browser tab
4. Confirm reconnect URL contains `?afterEventId=<cursor>`
5. Confirm no duplicate candles, no missing candles

9.4 Chart Authority Activation Readiness

Script: `bash scripts/chart_authority_activation_readiness.sh`

Required result: All 7 gates PASS

9.5 Parity Results

Command: `curl -s http://localhost:3000/api/market-data/parity | jq .`

Required result: `mismatchRate` ≤ 0.02 over ≥ 100 bars

Section 10 — Production Deployment Note

Gate G4 is NOT yet approved. `DATABENTO_CHART_AUTHORITY` is NOT yet activated.

When Gate G4 is approved by Phil in writing, the activation procedure is:

1. Run `bash scripts/chart_authority_activation_readiness.sh` — must return ALL PASS
2. Set `MARKET_DATA_AUTHORITY=DATABENTO_CHART_AUTHORITY` in the production `.env`
3. Set `ATLAS_GATE_G4_CHART_AUTHORITY_ENABLED=true` in the production `.env`
4. Restart the Atlas Nexus server
5. Verify the dashboard shows the “DATABENTO CHART AUTHORITY” status indicator
6. Monitor parity metrics for 30 minutes before considering the activation stable

Sprint 123A.5 will not begin until Phil explicitly approves Gate G4 in writing.

Document prepared by Atlas Nexus autonomous research engine. Gate G4 approval authority: Phil (human).